

Jacob Edwards, PhD

Data Scientist

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portfolio: <https://amnion.github.io/>

I have 13 years' experience building end-to-end data pipelines and statistical models for complex datasets. My work has focused on time-series and unstructured data, where I have developed scalable workflows, automated quality control systems, and predictive models that improve efficiency and enable decision-making. I bring a strong foundation in statistical inference, machine learning, and experimentation, along with a track record of translating technical results into actionable insights for diverse stakeholders.

SKILLS

Programming & Tools

Python, SQL, R, MATLAB, MS Office

Data Analysis

Statistical modeling, regression and classification, time series analysis, natural language processing, machine learning, feature engineering, dimensionality reduction, model evaluation and prediction, data cleaning, wrangling, and visualization

Experimentation

Hypothesis testing, experimental design, sampling methods, effect size estimation, bootstrapping, simulation

Communication

Data storytelling, technical writing, oral presentation, cross-functional collaboration, teaching and mentoring

RELEVANT EXPERIENCE

Research Scientist

COLUMBIA UNIVERSITY

New York, NY

2019 - 2025

Time-Series Modeling of Brain Data

- Generated and analyzed 3.2 TB of time-series neural data. Built scalable workflows for data cleaning, feature extraction, aggregation and visualization.
- Implemented methods to understand stimulus-response relationships, including kernel smoothing, adaptive thresholding for event detection, auto- and cross-correlation functions, and Fourier transforms.
- Replaced a manual data quality screening workflow with an automated, supervised ML-based [anomaly detection classifier](#), reducing manual review time by ~20% and improving consistency across projects.
- Built [interactive dashboards](#) to support exploratory analysis and enable non-technical collaborators to interpret multi-dimensional results.

Sequence Modeling of Vocal Acoustic Data

- Generated and analyzed datasets comprising 31,081 vocal acoustic events across 179 subjects. Optimized event segmentation pipeline to derive structured tabular representations from raw waveforms.
- Engineered features from audio recordings using domain metrics and decomposition techniques (PCA, wavelets) for downstream application of dimensionality reduction and cluster analysis to identify latent structure and correlations.
- Unified infrastructure for database management and data annotation by creating a [graphical user interface](#) that reduced work hours per project by ~80% and was used internally by 3 other teams.
- Developed and evaluated NLP model for sequence prediction using non-lexical signal features. Designed custom cost metric based on fuzzy matching algorithms. Created custom effect size metrics and bootstrapped confidence intervals.

Leadership and Cross-Functional Impact

- Secured \$140,256 in competitive [federal research funding](#) by translating technical proposals into articulated project aims and measurable outcomes.
- Developed and taught an original, [applied data science curriculum](#) (20 hours of lecture, 7 homework assignments) covering R programming, statistical modeling, data wrangling, visualization and machine learning. Emphasized translating real-world data analysis into actionable policy recommendations for legislators and decision-makers.
- Delivered presentations to technical and non-technical audiences via posters and seminars within the department, at national conferences and community events such as *Taste of Science NYC* and *BioBus Harlem*.
- Authored 6 peer-reviewed papers in high-impact journals such as *Nature Scientific Reports* and *Journal of Neuroscience*.

SELECTED PUBLICATIONS

2026	Models of species' syntax rules accurately predict vocal sequences	[paper]
2025	Temporal organization of behavior is driven by species identity	[paper]
2023	Machine learning analysis of song links vocal acoustics to ancestry	[paper]

EDUCATION

PhD	2025	Columbia University
MSc	2018	Colorado State University
BSc	2013	Uni. of Tennessee, Knoxville